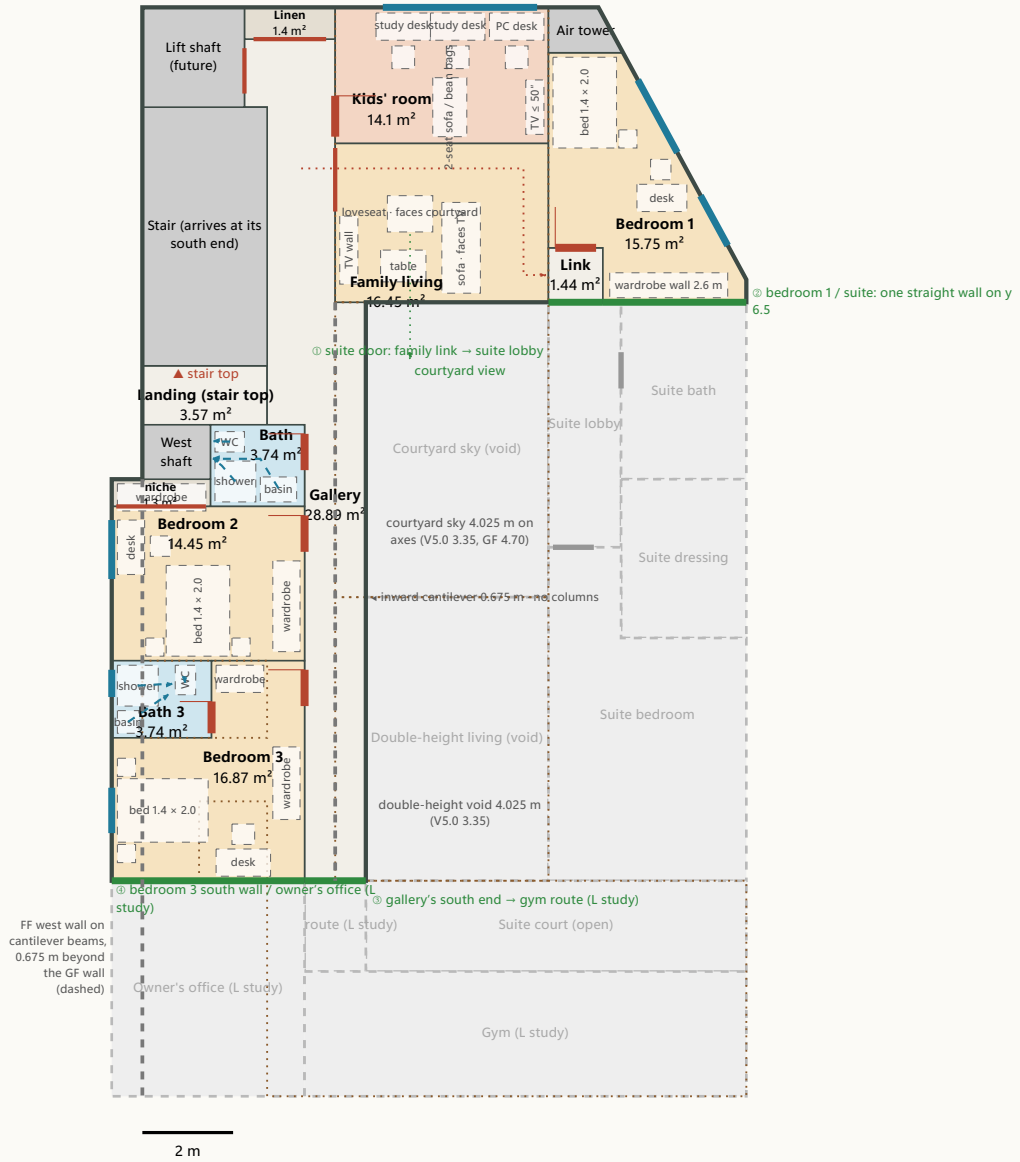
First floor · family part · focus study

Taif / Palm House · first floor V5.1 (split projection) · 24 Oct 2026 · local = main CSV coordinates (x east, y south) · wall centre lines · north up

First floor V5.1 · part 1 of 2: family part · projection split 0.675 m in + 0.675 m out, west rooms 4.25 m wide · part 2 (office, gym, suite court, master suite): studies/ff_suite.typ



Rooms

Room	Axes w × d (m)	Axes m²	≈ clear m²
Landing (stair top)	2.75 × 1.3	3.57	3.14
Gallery (north part + on the projection)	polygon	28.89	25.92
Linen cupboard	2 × 0.7	1.4	–
Family bath (bed 1 + 2, family living)	2.08 × 1.8	3.74	2.96
Wardrobe niche (bed 2)	2.17 × 0.6	1.3	–
Bedroom 2	4.25 × 3.4	14.45	12.84
Bedroom 3 bath (en-suite)	2.2 × 1.7	3.74	2.92
Bedroom 3	polygon	16.87	14.97
Kids' room (study + gaming)	4.7 × 3	14.1	12.44
Family living	4.7 × 3.5	16.45	15.16
Family link	1.2 × 1.2	1.44	1.08
Bedroom 1	polygon	15.75	13.83
Fixed: stair, lift, west shaft, air tower		23.8	
Zone		**145.51**	

Clear ≈ axes minus half a wall on every edge (250 mm outer, 200 mm inner, ≈ 50 mm balustrade on the void edges, nothing on open edges), corners ignored.

✓ All rooms inside the zone · no overlaps · no unassigned area · furniture inside its room, not overlapping, out of the routes · door swings clear · every room has a door reached from circulation · routes ≥ 1.0 m clear · suite door and gallery end on circulation

Key numbers

- **V5.1 · split projection:** the V5.0 1.35 m is split: **0.675 m inward** over the courtyard and the double-height void (x 4.25–4.93, a plain cantilever, **no columns**) and **0.675 m outward** beyond the GF west wall (x 0 → −0.68, y 10.4–24, the west wall on cantilever beams). The whole west band (rooms, gallery, office) moves 0.675 m west; stair, lift, landing and west shaft stay.
- **Courtyard sky** 4.70 (GF) → 3.35 (V5.0) → **4.025 m** on axes; **double-height void** the same. The ground-floor courtyard and living keep their full floor.
- **West rooms** still 4.25 m on axes (≈ 4.03 m clear). **Gallery on the projection:** 1.35 m, ≈ **1.23 m clear** (½ wall + ½ railing); 2.18 m wide at the stair top (y 6.5–9.2). **Circulation in the zone:** 33.9 m² (landing, gallery incl. the projection, family link); the walk through the living is not counted.
- **Family living:** 16.45 m², on the courtyard edge, no separate route. **Kids' room:** 14.1 m².
- **Bedrooms:** 2 · 14.45 m² (4.25 × 3.4) · 3 · 16.87 m² + its own bath 3.74 m² · 1 · 15.75 m² (no bath). **Family bath** 3.74 m² for bedrooms 1 and 2, the family living and the kids' room.
- **Walked to the family bath door:** bedroom 2 ≈ 2.7 m · family living ≈ 7.7 m · bedroom 1 ≈ 16.2 m. Bedroom 3 → its own bath ≈ 2.7 m.
- **Owner's office:** bedroom 3 has moved up into the west band; the freed south-west corner (≈ 16 m²) becomes the owner's office, in the L study (studies/ff_suite.typ), its door on the gym route.

Plan: red = doors (thin = opening without a leaf) with swing line · blue = windows · blue dashed = drains · green = interfaces · brown dotted = ground floor below (the full courtyard and living floors) · grey dashed = the two GF wall lines the cantilevers start from (x 4.25 courtyard edge, x 0 west wall) · grey = neighbours · dashed boxes = furniture and fixtures.

Family part · the west projection and the rooms

V5.1 · the 1.35 m projection split: 0.675 m in, 0.675 m out

- **The idea:** V5.0 put the whole 1.35 m over the courtyard and the double-height void (x 4.25–5.6, on three columns). V5.1 puts **half inward** (x 4.25–4.93, y 6.5–19.25) and **half outward** beyond the GF west wall (x 0 → −0.68, y 10.4–24). The whole west band (bedrooms 2 and 3, bedroom 3 bath, wardrobe niche, gallery, and in the L the office and the gym route) moves 0.675 m west; stair, lift, landing and west shaft stay where they are, so the west face steps out at y 10.4, just south of the shaft.
- **Split:** the band is 4.25 (rooms) + 1.35 (gallery) = 5.6 m and stays so; rooms ≥ 4.0 m clear needs ≥ 4.225 m on axes, the gallery ≥ 1.0 m clear needs ≥ 1.125 m. Any split works for the widths; the **site** limits the outward part to 0.785 m (the GF outer face is 0.785 m from the M line). The even split 0.675 / 0.675 leaves 0.11 m to the M line.
- **Inward part:** a plain **0.675 m cantilever** from the courtyard edge x 4.25 (back span 1.5 m to the GF riwaq wall x 2.75), carrying only the gallery and its railing: **no edge beam, no columns** (V5.0 had three on x 5.6).
- **Outward part:** the FF west wall stands on **cantilever beams** 0.675 m beyond the GF west wall, from y 10.4 to 24 (13.6 m), back span 2.75 m to the GF wall x 2.75. It shades the GF west wall from the afternoon sun and roofs the guest entrance (foyer door, y 21.5–22.9).
- **Before / after:** courtyard sky 4.70 (GF) → 3.35 (V5.0) → **4.025 m** on axes; double-height void the same; only x 4.25–4.93 of the GF has the lower ceiling.
- **Gallery on it:** 1.35 m (≈ 1.23 m clear) with a railing on the void edge, 2.18 m wide at the stair top; every west room door opens from it; no room opens onto the stair landing.

Kids' room · family living · bedroom 1 (unchanged)

- **Kids' room** 4.7 × 3.0 m: two study desks + PC desk under the north window, gaming corner (TV ≤ 50" on the east wall, ≈ 1.3 m viewing distance), solid door from the gallery.
- **Family living** 4.7 × 3.5 m: a view, not a path; the walk to the family link passes behind and beside the sofa.
- **Bedroom 1:** one room, no bath, its south wall straight on y 6.5; it uses the family bath.

Family bath · 2.075 × 1.8 m, beside the west shaft (variant: was 2.75)

- WC on the shaft wall, shower 0.9 × 0.9, basin by the door; door from the gallery on the projection (opening in, swing clear), not onto the landing. For bedrooms 1 and 2, the family living and the kids' room.
- **Forced by the shift:** its west wall is the fixed shaft (x 1.5) and its east wall follows the band to x 3.575, so it loses 0.675 m (3.74 m², was 4.95). Compact but complete; the basin moves 0.2 m west.

Bedroom 2 · 4.25 × 3.4 m (≈ 4.0 × 3.2 clear) + wardrobe niche 2.175 m

- **Bed 1.4 × 2.0** with its head on the south wall and **bedside tables on both sides**; **wardrobes** in the niche beside the shaft (now 2.175 m wide: it gains the 0.675 m outside the shaft) and on the east wall (1.6 m); **desk** under the west window. Door from the gallery (swing clear), ≈ 2.7 m from the family bath.

Bedroom 3 · 16.87 m² + its own bath 2.2 × 1.7 m

- **Now in the west band** (it was on the men's majlis roof): an entry with a wardrobe, then the room 4.25 × 3.15 m: **bed 1.4 × 2.0** head on the west wall with two bedside tables, wardrobe on the east wall, desk on the south wall. Door from the gallery (swing clear).
- **Own bath (en-suite)** off its entry: shower, WC, basin; door swing clear; ≈ 2.7 m from the bedroom door. It stands **over the GF family WC** (walls on the GF lines y 14.4 and 16.1; the WC is still over the GF WC, x 0–2.75); variant: its west 0.675 m (shower, basin) is on the outward cantilever, the drains run in the floor to the WC stack.

Gym route and the owner's office → the L study

- The gallery ends at y 19.25 (x 3.575–4.925); from there the **gym route** and the **owner's office** belong to the L study (studies/ff_suite.typ), with the gym, the court and the suite. Variant: both move 0.675 m west with the band.

Interfaces with the master suite study (fixed)

- ① suite door at the end of the family link (y 6.5); ② bedroom 1 / suite: one straight wall on y 6.5; ③ the gallery's south end (y 19.25, x 3.575–4.925), where the gym route starts; ④ bedroom 3's south wall (y 19.25), now a wall to the owner's office (no window).

Family part · routes, walls, drains, changes, open items

Routes (≥ 1.0 m clear)

Route	Axes	≈ clear	Deducted
Landing	1.3	1.2	½ wall on the south side; the north side is the stair arrival
Gallery north (beside the stair)	1.5	1.3	½ wall each side
Gallery on the projection	1.35	1.22	½ wall (rooms side) + ½ railing (void side)
Walk behind the sofa (in the living)	1.1	1	½ wall on the kids' room side; the sofa side is open
Walk beside the sofa to the family link	1.1	1	½ wall on the bedroom 1 side; the sofa side is open
Family link	1.2	1	½ wall each side (bedroom 1 to the north and east)

Walking distance	m
Stair top → kids' room door	8.8
Stair top → family living (opening)	8.7
Stair top → bedroom 1 door	15.9
Stair top → ⓘ suite door	15.9
Stair top → family bath door	5
Stair top → bedroom 2 door	6.8
Stair top → bedroom 3 door	10.2
Stair top → gallery's south end (L)	13.8
Bedroom 2 door → family bath door	2.7
Family living → family bath door	7.7
Bedroom 3 door → its own bath door	2.7
Bedroom 1 door → family bath door	16.2

Stair top → every room, the suite door and the gallery's south end runs only on the landing, the gallery (north part and on the projection), the family link and the walking line through the living (checked: every door's approach is on circulation, no furniture in a route).

Engineer

- **Inward cantilever (V5.1):** 0.675 m slab beyond the courtyard edge x 4.25, y 6.5–19.25, back span 1.5 m to the GF riwaq wall x 2.75; it carries only the gallery (live load) and its railing. A cantilever / back-span ratio of 0.45 and a short reach: **no columns, no edge beam** expected; the x 4.25 beam line already carries the rooms' wall in V5.0. Engineer to confirm slab depth, the x 4.25 beam and the railing fixing.
- **Outward cantilever (V5.1):** the FF west wall (250 mm, ≈ 3.6 m high, + the parapet) stands 0.675 m beyond the GF west wall, y 10.4–24: **wall on cantilever beams** (one at each cross wall, back span 2.75 m to x 2.75, with an edge beam under the wall). Engineer: beam depth, line load of the wall, tip deflection (a cracking limit for the wall), the corner at (−0.68, 24).
- **Ground floor changes only by:** removing the three V5.0 columns (none needed), the lower ceiling under the inward 0.675 m only, and the cantilever beams' fixings in the west wall.

Walls off the ground-floor lines

Ground-floor lines here: x 0, 1.25 (guest WC), 1.5 (shaft), 2.25, 2.75, 4.25, 8.95 · y 2.2, 7.9, 10.4, 12.9, 14.4, 16.1, 17.5, 19.25, 21.4 (foyer) (west band); the women's majlis (x 4.25–8.95, y 0–6.5) and the men's majlis (x 2.75–13.311, y 19.25–24) have no inner wall; east part (bedroom 1, family link): see the V4.1 list.

- **Off:** y 3.0 (kids' room / living, over the women's majlis); y 11.0 (family bath and niche / bedroom 2) and x 1.5 south of the shaft (niche); **x 3.575** (the rooms' east wall, 0.675 m west of the GF courtyard edge, on the slab between x 2.75 and 4.25); **x −0.68** (the west wall, on the cantilever beams); x 1.525 (bedroom 3 bath / entry); (the gym route walls: see the L study).
- **On:** y 14.4 and 16.1 (bedroom 3 bath = the GF family WC walls), x 1.5 / y 10.4 (shaft), y 6.5 (bedroom 1 / suite), the outer walls north of y 10.4.
- **Gone upstairs:** the wall on x 2.75 south of the landing (the rooms span over the GF riwaq); the GF riwaq wall x 2.75 below carries the floor as before.

Drains (blue dashed)

- **Family bath:** WC on the west shaft wall (≈ 0.3 m), shower ≈ 0.5 m, basin ≈ 1.8 m, in the bath floor to the shaft; floor over the GF utility room and hospitality kitchen, nothing in their ceilings.
- **Bedroom 3 bath:** over the GF family WC: WC straight into its stack, shower ≈ 1.0 m, basin ≈ 1.1 m, both on the outward cantilever (drains in the floor build-up, nothing below but the cantilever slab). Wet room on a wet room.
- **Bedroom 1:** no bath. The east duct is removed (suite bath to a stack outside the east wall, see ff_suite.typ).

What changed in V5.1 (24 Oct 2026, from V5.0)

- **Projection split** 0.675 in + 0.675 out; the band moves 0.675 m west (y 10.4–24); courtyard sky and void 4.025 m (V5.0 3.35); **no columns**.
- **Forced:** family bath 2.75 → 2.075 m wide (shaft fixed); wardrobe niche 1.5 → 2.175 m; gallery 2.18 m wide at the stair top; bedroom 3 bath wall x 2.2 → 1.525; all west doors, windows and furniture 0.675 m west.
- **Unchanged:** kids' room, family living, family link, bedroom 1, landing, linen, stair, lift, west shaft; bedrooms 2 and 3 keep their size.

Open items

- **TOP OPEN ITEM (before construction drawings):** the FF west face is 0.11 m from the M line. Municipality / engineering office to confirm that an upper-floor projection this close to the M line is allowed; if not, back to the V5.0 solution (1.35 m inward on three columns at x 5.6, tag v5.0).
- **Built-in cupboards in niches** (linen cupboard 2.0 m, bedroom 2 wardrobe niche 2.175 m): both have walls on both sides. Prefer a **joinery box** (own 18 mm sides, top and frame, scribed to the walls). Decide with the carpenter (docs/DETAILS.md).
- **Engineer:** the outward wall on cantilever beams; the inward 0.675 m cantilever without columns.
- **Courtyard light:** sky 4.025 m between ≈ 7–8 m walls; to check with a daylight study.